

Dharun Pranaav K B

Electronics & Communication Engineering · PES University

dpranaav@gmail.com | 9489714872 | Bengaluru, India

[GitHub](#) | [LinkedIn](#)



Aspiring Data Scientist with experience in machine learning, computer vision, time-series forecasting, and analytics-driven software development. Published IEEE researcher, ISRO intern, and Software Engineering intern with strong Python, SQL, Pandas, and statistical modeling skills.

EDUCATION

PES University, Bengaluru, India

B.Tech, Electronics and Communication Engineering

Sep 2022 – May 2026

GPA: 7.54

Advaith International Academy

XII – CBSE Board

86%

WORK EXPERIENCE

Software Engineering Intern

Jan 2026 – May 2026

DUXES Labs, Bengaluru

- Designed and developed **VirtualLab**, a full-stack remote lab access platform enabling browser-based remote desktop sessions on physical lab hardware.
- Built a **Node.js/TypeScript** backend with Express and Prisma (PostgreSQL) handling booking lifecycle, JWT authentication, and an async task queue.
- Developed a **React** frontend with Firebase Auth and a real-time RDP readiness polling system.
- Engineered a **Python connector service** for automated MeshCentral CLI orchestration, guest share provisioning, and email delivery via Gmail SMTP.
- Integrated **Cloudflare Tunnel** for secure external access without inbound firewall exposure.

Research Intern

July 2025 – Aug 2025

U R Rao Satellite Centre (ISRO), Bengaluru

- Developed an end-to-end satellite image processing pipeline for the **SpaDeX mission**.
- Handled multi-channel raw sensor data through **CCSDS 122B** decompression, checksum validation, and channel reconstruction.
- Applied image processing techniques including demosaicing, white balancing, histogram correction, and post-processing to produce high-fidelity **PNG/TIFF** outputs.
- Navigated real-world constraints of space imaging data from an active Indian Space Research Organisation mission.

PUBLICATIONS & PRESENTATIONS

Image Captioning using Xception and LSTM with T5 Refinement

17th IEEE International Conference on Computational Intelligence and Communication Networks (CICN 2025)

[IEEE Xplore](#) ↗

SAP Flow Forecasting using SARIMAX with Diurnal Exogenous Simulation

International Conference on Renewable Energy Systems (ICRES 2026)

Accepted – Upcoming Presentation

PROJECTS

Image Captioning Model

Published at CICON 2025 — [IEEE Xplore](#) ↗

Built an end-to-end image captioning pipeline using **Xception** for CNN-based visual feature extraction and an **LSTM** decoder with beam search for sequence generation. Applied a **T5-based** language refinement module to improve grammatical accuracy and fluency of generated captions. Processed and analyzed the Flickr8k dataset using Python, Pandas, TensorFlow and Keras. Trained and evaluated the model using BLEU-based metrics. Work accepted and presented at an IEEE international conference.

Football Scouting Algorithm

Built a data-driven player recommendation engine using FIFA, FBref, and Wyscout datasets containing player and match statistics. Performed data cleaning, feature engineering, and exploratory analysis using Python, Pandas, and SQL. Applied K-Means clustering, Random Forest, and Logistic Regression models to identify statistically similar replacement candidates and rank recommendations using a custom chemistry scoring framework.

SAP Flow Forecasting System

Accepted at ICRES 2026

Designed and trained a SARIMAX forecasting model on five years of environmental data to predict hourly sap flow trends. Performed feature engineering, statistical analysis, and time-series modeling using temperature, humidity, and AQI variables to generate 7-day forecasts with confidence intervals. Built an interactive dashboard for real-time monitoring with uncertainty intervals. Deployed on **NVIDIA Jetson Nano** with Arduino-based sensor integration for continuous data acquisition.

SKILLS

Programming	Python, SQL, C, MATLAB, R (Learning), SAS (Learning)
Data Science	Pandas, NumPy, Scikit-learn, TensorFlow, Keras, Feature Engineering, Statistical Analysis, Time Series Forecasting
ML / AI	Regression, Classification, Clustering, Computer Vision, Deep Learning, Model Evaluation
Databases	PostgreSQL, Firebase, Prisma ORM
Web / Backend	Node.js, TypeScript, React, Express, REST APIs, JWT Authentication
Tools	Git, OpenCV, Vivado, CST, RIPES, Wireshark, Arduino, NVIDIA Jetson Nano
Concepts	Data Structures & Algorithms, Image Processing, Forecasting, Database Design